IndianOil	**LSTK PACKAGE-2 OFFSITE AND UTILITIES FOR "ETHYLENE GLYCOL" AT PARADIP REFINERY, ODISHA**	L&T Hydrocarbon Engineering
TOYO ENGINEERING	**Doc. No.** : IPEG-6389-8830-QA-002-0006 **Title** : Positive Material Identification Procedure	**Rev. No.** : A1

OWNER : **INDIAN OIL CORPORATION LIMITED (IOCL)**

PMC : **TOYO ENGINEERING INDIA PRIVATE LIMITED (TEIPL)**

LSTK CONTRACTOR : **L&T HYDROCARBON ENGINEERING LIMITED (LTHE)**

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Name	Discipline	Sign	Date

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Rev.	Description	Date	Prepared By	Reviewed By	Approved By
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CHANGE LIST

SL. NO.	REV. NO.	PAGE NO.	CLAUSE / LINE NO.	BRIEF DESCRIPTION / REASON OF REVISION	REMARKS
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1 PURPOSE

1.1 The purpose of this document is to provide requirements of Positive Material Identification (PMI) activities to be followed during procurement of Equipment, Packages and Materials (E&I, Piping etc.) included in LSTK Package-2 Offsite and Utilities for "Ethylene Glycol" at Paradip Refinery, Odisha.

2 SCOPE

2.1 This specification applies to the requirements for Positive Material Identification (PMI) to be performed at the Vendor's / Sub Vendor's works on all Metallic Alloy Materials (Ferrous and Non Ferrous) and Welds procured either directly by LTHE / Vendor or indirectly through the sub-Vendor.

2.2 This Procedure shall be followed by all LTHE, Vendor and Sub Vendor of order placed by LTHE for LSTK Package-2 Offsite and Utilities for "Ethylene Glycol" at Paradip Refinery, Odisha, India.

2.3 This Specification covers the procedures and methodology to be adopted to assure that the chemical composition of following Metallic Alloy Material and Weld is consistent with the material specifications as specified in purchase documents using 'Alloy Analyzer' at the time of final inspection before dispatch.

- Low Alloy Steels
- Stainless Steels
- Nickel Alloys
- Copper Alloys
- Titanium Alloys

2.4 The scope of this specification shall include but shall not be limited to PMI to be performed on Metallic Alloy Material as listed below.

- Bulk Piping Material as Pipes, Fittings, Flanges, Valves, Fasteners etc.
- Gaskets (for ring type joints)
- Instrumentation items as Control valves, Safety valves, Tubing for corrosion purpose (i.e. Inconel), level displacer and float switches (Chamber and Head), In-line Instruments, etc.
- Materials for fabricated equipment & rotating equipment
- Welds in equipment, piping etc.

2.5 Following items shall be excluded from the scope of PMI examination

- Gaskets other than for Ring Type Joints
- Internal components of Valves

- Internal Instrument parts
- Internal Machinery Parts
- Electrical Components
- Rotating Equipment internals
- Machinery lube oil, cooling water and steam piping
- Non-Pressure retaining parts and welds (except those directly attached to pressure containing components)
- Compression-type ferrules

2.6 All grades of Metallic alloy materials & welds shall be subject to a PMI test at site. In case of any defective materials being found at site, the Vendor shall be responsible to effect replacement of such defective materials at project site without any delays to the satisfaction of PMC / Owner.

3 DEFINITION

Term	Definitions
OWNER / CLIENT	**INDIAN OIL CORPORATION LIMITED, PARADIP or Their Authorized Agency**
PMC	**TOYO ENGINEERING INDIA PVT. LTD**
LSTK CONTRACTOR	**LTHE Hydrocarbon Engineering Limited (LTHE)** The Person(s), Firm, Company, Organization responsible for executing the specified work inclusive of Engineering, Procurement, Construction and Commissioning on lump sum turnkey basis.
VENDOR	Any firm, company, organization on whom an order is placed by LTHE for the supply of referred items.This definition shall also include any sub Vendor or manufacturer on whom a sub order is placed by the Vendor.
ALLOY MATERIAL	Any metallic material of Ferrous or Non Ferrous grade (including welding filler material) that contains alloying elements such as chromium, nickel, molybdenum, vanadium etc., which are intentionally added to enhance mechanical or physical properties and / or corrosion resistance.

INSPECTION LOT	A group of items offered for inspection covered under the same size, heat and heat treatment lot.
SAMPLE	One or more units selected at random from the Lot which are to be examined to determine acceptability of Lot.
IN LINE INSTRUMENTS	Instruments installed in the process line, e.g. flow meter, thermowell, rupture disc, level displacer, etc. This excludes On-line instruments with direct/open connections to the process, installed remote from the process line, e.g. transmitters, gauges.
PMI	The term Positive Material Identification (PMI) refers to determination / verification of alloy type or its composition using portable or mobile spectrometer / alloy analyzer.
TPIA	LTHE appointed Inspection Agency

4 REFERENCE

4.1 IEGP-6351-8640-TP-000-002- Positive Material Identification Procedure for LSTK contractor, vendor or sub-vendor.

4.2 PDRP-8440-SP-0014 – Positive Material Identification (PMI) of Bulk Alloy Materials.

4.3 PDRP-8440-SP-0015 – Positive Material Identification (PMI) at Supplier Work.

4.4 API recommended practices 578 (Third Edition, February 2018) - Material Verification Program for New and Existing Alloy Piping Systems.

5 EQUIPMENT

5.1 The acceptable instruments for alloy analyzer shall be either "Portable X-Ray fluorescence" or "Optical Emission" type each capable of verifying the percentage of alloy elements within specified range. The following methods / instruments are acceptable.

(a) Portable X-Ray Fluorescence Analyzers TN Technologies Alloy Analyzer 9266, 9277 (the metallurgist XR) or Metallurgist Pro, Metorex – X – MET 880, X-MET 960 or X-MET 2000 or Equivalent and any upgraded Model shall be acceptable subject to satisfactory demonstration of calibration and accuracy at the time of testing and approval by PMC/OWNER.

(b) Portable Optical Emission Analyzer Spectro port model TP-07 or TFO-02, Spectro Test F, Metorex ARC-MET 900, or ARC-MET 930 or Equivalent and any upgraded Model shall be acceptable subject to satisfactory demonstration of calibration and accuracy at the time of testing and approval by PMC/OWNER.

6 CALIBRATION

6.1 The PMI Instrument used shall have the sensitivity to detect the Alloying Elements in the specified range.

6.2 All PMI instruments shall have been serviced within six-month period of the time of use to verify the suitability of batteries, sources etc., and the data of the last service shall be stated on the PMI Report.

6.3 Each analyzer must be calibrated according to the manufacturer's specification at the beginning and end of each shift. Instrument must be checked against known standard for each alloy type to be inspected during the shift.

6.4 Certified samples with full traceability of known alloy materials shall be available for use as a random spot check on the instrument calibration.

7 PMI EXAMINATION

7.1 PMI examination of alloy materials and welds is independent of any certification, markings or color coding that may exist and is aimed at verifying that the alloy used are as per specified grades.

7.2 Vendor or Sub vendor shall identify all incoming alloy materials and maintain full traceability of all alloy materials, including all off-cuts. Transfer of identification marks shall be undertaken prior to cutting, to ensure maintenance of identification on off cuts.

7.3 Vendor or Sub vendor shall ensure that all alloy materials are segregated & stored in separately identified locations to prevent mix up of materials of different alloy specifications or Alloy materials with carbon steel. Non Ferro magnetic materials shall be segregated at all times from Ferro magnetic materials.

7.4 PMI Examination shall be subject to witness inspection by LTHE/TPIA/PMC/OWNER as per approved ITP (Inspection and Test Plan)

7.5 For Equipment, PMI shall be performed on completed or assembled parts of equipment before Hydro Test / Pneumatic Test / Factory Acceptance Test etc., as applicable. Weld joints shall

be tested after final acceptance of applicable NDT. PMI shall be repeated in case repairs carried on weld joint after PMI testing and shall be recorded.

7.6 The PMI shall be done earlier, if accessibility problem is expected due to sequence of work, or on the other hand, acceptance shall be based on material / weld traceability and certification if the item to be tested is inaccessible due to configuration.

8 ACCEPTABLE METHOD OF PMI

8.1 The method used for PMI examination shall provide a quantitative determination of the alloying elements like chromium, molybdenum, nickel, vanadium etc. in metallic alloy materials.

8.2 Instruments or methods used for PMI examination shall be able to provide quantitative, recordable, elemental composition results for positive identification of alloying elements present.

8.3 Chemical spot testing, magnets, alloy sorters and other methods using eddy current or triboelectric testing methods are not acceptable for PMI examination.

8.4 The surfaces to be examined shall be prepared by light grinding or abrasive paper and solvent cleaner. Evidence of arc burn resulting from examination shall be removed by light grinding or abrasive paper. No permanent marks, which are injurious to the usage of product in service, are acceptable.

8.5 Ring Type Joint gasket of alloy material shall be inspected by using portable X-Ray Fluorescence Instrument.

8.6 Testing shall be done as per procedures outlined by the manufacturer of alloy analyzer being used. Modification of these procedures if any must be approved by the manufacturer of the Alloy analyzer.

9 PERSONNEL QUALIFICATION

9.1 The persons performing the PMI test should have sufficient knowledge about properties of material, all aspects of operation of PMI equipment including the method of testing.

9.2 The persons performing PMI shall demonstrate their capabilities to the satisfaction of the visiting TPIA representative.

10 EXTENT OF PMI EXAMINATION

10.1 Following sampling plans shall be applicable for PMI examination of various alloy material items.

A. Flanges, Fittings, Valves, RTJ Gaskets – 100%

B. Pipes / Tubes – 100% (Procured from traders)

C. Pipes / Tubes – 2 random samples drawn from each Size/Heat/Lot (Procured directly from mills)

D. Fasteners – Refer below table

Lot Size	Sample Size
Up to 100	2% (Min 2 Nos.)
101 to 500	1% (Min 3 Nos.)
501 and above	0.5% (Min 5 Nos.)

E. Fabricated/ Static/ Rotating Equipment – All Pressure retaining parts & welds

F. Instruments – 10% for each type, size and rating; 100% for Valves

Notes:

1) For Welded Pipes and Fittings, PMI shall be performed on Base Metal as well as weldments.
2) Whenever any sample drawn to PMI test on the basis of percentage selection in C, D & F above, fails to meet specification requirements, 100% of items of lot shall be tested for PMI.
3) Any Failure thereafter during check as per Note 2 above, shall mean rejection of whole lot.

11 ACCEPTANCE CRITERIA

11.1 PMI acceptance criteria for materials / welding consumable / welds will be as per relevant codes / standard like ASTM/ASME/AWS etc.

12 REJECTED MATERIALS

12.1 Whenever material is identified as rejected, the material shall be suitably marked with a red "X" using indelible, non-deleterious ink or paint.

12.2 Rejected material shall be segregated in a separate, designated area, to prevent their re-use.

12.3 Rejected weld metal shall be removed by suitable means and the excavation subject to a PMI check prior to repair. Alternatively, the weld shall be completely removed, the weld preparation re-made and surface Non Destructively Examined, and re-welded. All repair welds shall be subject to PMI.

12.4 Vendor shall have the option of performing a more accurate analysis on rejected material, at his own care and cost, to verify the composition of the rejected material.

12.5 Vendor shall be responsible for replacing all rejected material with the correct specified material and performing PMI on this replacement material.

13 MARKING

13.1 All alloy materials tested by PMI shall be identified using either of the following methods by indicating "AV".

- Bar Code/Hologram Sticker
- A low stress stamp marking

Above marking shall be traceable with Test Certificate / PMI Report.

14 RECORDING AND DOCUMENTATION

14.1 The results of PMI examination shall be recorded in a Report Format as enclosed with this procedure.

15 SAMPLE FORMAT

POSITIVE MATERIAL IDENTIFICATION REPORT									Page ____of____
Project: Ethylene Glycol Project	Client : IOCL								Job No.
Vendor / Sub Vendor Name:									
PMI Report No. :	Supplier/Sub-Supplier :								
PMI Procedure No. with Rev.:	Testing Agency :								
Purchase Order No. :	PMI Location:								
Bulk Item Type (as per Requisition)									
Material Specification / Grade									
Number of items in Lot									
Requisition Item No. / Description	Alloy Content, Weight Percent								Remarks
Element*	**Cr**	**Ni**	**Mo**	**V**	**Cu**	**Ti**	**Nb**	**Ta**	Accept / Reject
Specified Range									
Actual Observations 1.									
2.									
3.									
4.									
5.									
Instrument Type / ID:					Last Service Date:				
Inspection Agency	Supplier QC						Witnessed By		

***** Elements to be analyzed/reported for positive material identification shall be as per TABLE 1 (Elements to be determine during PMI).

TABLE: 1 ELEMENTS TO BE DETERMINED DURING PMI

MATERIALS (1), (2)	**ELEMENTS TO BE DETERMINE**
Grade-12: 1 Cr - 0.5 Mo	Cr, Mo
Grade 11: 1.25 Cr - 0.5 Mo	Cr, Mo
Grade 22, 22L :2.25 Cr - 1 Mo	Cr, Mo
Grade 21, 21L: 3 Cr – 1 Mo	Cr, Mo
Grade 5: 5 Cr - 0.5 Mo	Cr, Mo
Grade 9: 9 Cr - 1 Mo	Cr, Mo, V
Grade 91: 9Cr – 1 Mo - V	Cr, Mo, V
12 Cr (Type 410S/405)	Cr
12 Cr	Cr
17 Cr	Cr
304 (L)	Cr, Ni, Mo, Nb(Cb), Ti
310	Cr, Ni, Mo, Nb(Cb), Ti
309 (L)	Cr, Ni, Mo, Nb(Cb), Ti
309 Nb	Cr, Ni, Mo, Nb(Cb), Ti
316 (L)	Cr, Ni, Mo, Nb(Cb), Ti
321	Cr, Ni, Ti, Nb(Cb)
347	Cr, Ni, Nb(Cb), Ti
Inconel 182/82	Ni, Cr
Inconel 625	Ni, Cr, Mo, Nb(Cb) + Ta, Ti
Inconel 600	Ni, Cr
Incoloy 800	Cr, Ni, Al, Ti, Cu
Incoloy 825	Cr, Ni, Mo, Ti
Admiralty Brass	Cu, Sn, As
Aluminium Brass	Cu, Al, Zn
Cupro-nickel (70-30)	Cu, Ni
Cupro-nickel (90-10)	Cu, Ni
Monel 400	Cu, Ni
Titanium	Ti

Notes:

(1) List of materials is not exhaustive, and shall not be construed as limiting the alloy materials subject to PMI.

(2) For Chemical elements of chemistries/Alloy grade that require PMI, and not addressed specifically in the above table, Vendor/Sub-vendor shall discuss the same with LTHE.